

3.1. CSM Experiment Dataset

In this research, the dataset includes 32 benchmark cases from published studies, five randomly generated synthetic instances, and five real-world cases from company XYZ. The table below summarizes each dataset. For example, instance N32 features four supplies and five demands (4×5 TP matrix) with an optimal cost of 1780. Optimal solutions were achieved using software named TORA (Taha, 2007)

1. Data sourced from 32 sets of journals can be seen in Table 3.1 below:

Table 3.1 Datasets from Journals

No	Reference	TP Name	TP Size	Optimal Value	Matrix
1	(Ramadan & Ramadan, 2012)	N01	3x3	5600	[Cij]3x3 = [32 40 120; 60 68 104; 200 80 60] [Si]3x1 = [20 30 45] [Dj]1x3 = [30 35 30]
2	(Samuel, 2012)	N02	3x4	28	[Cij]3x4 = [1 2 3 4; 4 3 2 0; 0 2 2 1] [Si]3x1 = [6 8 10] [Dj]1x3 = [4 6 8 6]
3	(Karagul & Sahin, 2020)	N03	5x5	1102	Cij]5x5 = [73 40 9 79 20; 62 93 96 8 13; 96 65 80 50 65; 57 58 29 12 87; 56 23 87 18 1] [Si]5x1 = [8 7 9 3 5] [Dj]1x5 = [6 8 10 4 4]
4	(Kaur et al., 2018)	N04	3x5	1580	[Cij]3x5 = [6 4 4 7 5; 5 6 7 4 8; 3 4 6 3 4] [Si]3x1 = [100, 125, 175] [Dj]1x5 = [60, 80, 85, 105, 70]
5	(Ahmed et al., 2016)	N05	4x4	410	[Cij]4x4 = [7 5 9 11; 4 3 8 6; 3 8 10 5; 2 6 7 3] [Si]4x1 = [30, 25, 20, 15] [Dj]1x4 = [30, 30, 20, 10]
6	(Ahmed et al., 2016)	N06	3x4	2850	[Cij]3x4 = [3 1 7 4; 2 6 5 9; 8 3 3 2] [Si]3x1 = [300, 400, 500] [Dj]1x4 = [250, 350, 400, 200]
7	(Das et al., 2014)	N07	3x4	1160	[Cij]3x4 = [6 1 9 3; 11 5 2 8; 10 12 4 7] [Si]3x1 = [70, 55, 90] [Dj]1x4 = [85, 35, 50, 45]
8	(Murugesan & Esakkiammal, 2020)	N09	4x4	111	[Cij]4x4 = [4 6 5 2; 6 4 1 4; 5 2 3 1; 4 6 7 8] [Si]4x1 = [6 10 12 14] [Dj]1x4 = [9 16 10 7]
9	(Amaliah et al., 2019)	N10	3x4	910	[Cij]3x4 = [20 2 20 11; 24 7 9 20; 8 14 16 18] [Si]3x1 = [30, 50, 20] [Dj]1x4 = [10, 30, 30, 30]

No	Reference	TP Name	TP Size	Optimal Value	Matrix
10	(Amaliah et al., 2019)	N11	3x4	2460	$[C_{ij}]_{3 \times 4} = [10 \ 2 \ 20 \ 22; 12 \ 7 \ 9 \ 40; 4 \ 14 \ 16 \ 32]$ $[S_i]_{3 \times 1} = [60, 100, 40]$ $[D_j]_{1 \times 4} = [20, 60, 60, 60]$
11	(Amaliah et al., 2019)	N12	3x3	425	$[C_{ij}]_{3 \times 3} = [15 \ 7 \ 25; 8 \ 12 \ 14; 17 \ 19 \ 21]$ $[S_i]_{3 \times 1} = [12 \ 17 \ 7]$ $[D_j]_{1 \times 3} = [12 \ 10 \ 14]$
12	(Juman & Hoque, 2015)	N13	3x3	4525	$[C_{ij}]_{3 \times 3} = [6 \ 8 \ 10; 7 \ 11 \ 11; 4 \ 5 \ 12]$ $[S_i]_{3 \times 1} = [150, 175, 275]$ $[D_j]_{1 \times 3} = [200, 100, 300]$
13	(Juman & Hoque, 2015)	N14	3x4	920	$C_{ij}]_{3 \times 4} = [4 \ 6 \ 8 \ 8; 6 \ 8 \ 6 \ 7; 5 \ 7 \ 6 \ 8]$ $[S_i]_{3 \times 1} = [40, 60, 50]$ $[D_j]_{1 \times 4} = [20, 30, 50, 50]$
14	(Ahmed et al., 2016)	N15	3x4	156	$[C_{ij}]_{3 \times 4} = [4 \ 6 \ 9 \ 5; 2 \ 6 \ 4 \ 1; 5 \ 7 \ 2 \ 9]$ $[S_i]_{3 \times 1} = [16 \ 12 \ 15]$ $[D_i]_{1 \times 4} = [12 \ 14 \ 9 \ 8]$
15	(Amreen, 2024)	N16	3x3	1510	$[C_{ij}]_{3 \times 4} = [22 \ 17 \ 25 \ 15; 13 \ 10 \ 16 \ 20; 16 \ 19 \ 9 \ 5]$ $[S_i]_{3 \times 1} = [30 \ 50 \ 50]$ $[D_i]_{1 \times 3} = [30 \ 30 \ 50 \ 20]$
16	(Gupta, 2020)	N17	3x4	743	$[C_{ij}]_{3 \times 4} = [19 \ 30 \ 50 \ 10; 70 \ 30 \ 40 \ 60; 40 \ 8 \ 70 \ 20]$ $[S_i]_{3 \times 1} = [7 \ 9 \ 18]$ $[D_i]_{1 \times 4} = [5 \ 8 \ 7 \ 14]$
17	(Amreen & Venkateswarlu, 2024)	N18	3x3	29	$[C_{ij}]_{3 \times 3} = [2 \ 5 \ 4; 6 \ 1 \ 2; 4 \ 5 \ 2]$ $[S_i]_{3 \times 1} = [4 \ 6 \ 6]$ $[D_i]_{1 \times 3} = [3 \ 7 \ 6]$
18	(Amreen, 2024)	N21	3x3	4205	$[C_{ij}]_{3 \times 3} = [135 \ 71 \ 260; 81 \ 126 \ 144; 170 \ 189 \ 201]$ $[S_i]_{3 \times 1} = [12 \ 17 \ 7]$ $[D_i]_{1 \times 3} = [12 \ 10 \ 14]$
19	(Rafi & Islam, 2020)	N22	3x4	12075	$[C_{ij}]_{3 \times 4} = [11 \ 13 \ 17 \ 14; 16 \ 18 \ 14 \ 10; 21 \ 24 \ 13 \ 10]$ $[S_i]_{3 \times 1} = [250 \ 300 \ 400]$ $[D_j]_{1 \times 4} = [200 \ 225 \ 275 \ 250]$
20	(Kaur et al., 2018)	N23	3x4	23	$[C_{ij}]_{3 \times 4} = [7 \ 2 \ 3 \ 7; 1 \ 5 \ 4 \ 5; 7 \ 2 \ 4 \ 5]$ $[S_i]_{3 \times 1} = [2 \ 1 \ 3]$ $[D_j]_{1 \times 4} = [2 \ 1 \ 1 \ 2]$
21	(Juman & Hoque, 2015)	N24	3x4	3460	$[C_{ij}]_{3 \times 4} = [20 \ 22 \ 17 \ 4; 24 \ 37 \ 9 \ 7; 32 \ 37 \ 20 \ 15]$ $[S_i]_{3 \times 1} = [120, 70, 50]$ $[D_j]_{1 \times 4} = [60, 40, 30, 110]$

No	Reference	TP Name	TP Size	Optimal Value	Matrix
22	(Juman & Hoque, 2015)	N25	3x4	809	[Cij]3x4 = [19 30 50 12; 70 30 40 60; 40 10 60 20] [Si]3x1 = [7 10 18] [Dj]1x4 = [5 7 8 15]
23	(G.Patel et al., 2017)	N26	4x5	490	[Cij]4x5 = [20 4 6 8 9; 4 10 8 18 8; 9 11 20 40 6; 7 6 9 14 16] [Si]4x1 = [20 30 15 13] [Dj]1x5 = [40 6 8 18 6]
24	(Korukoğlu & Balli, 2011)	N27	5x5	59356	[Cij]5x5=[46 74 9 28 99; 12 75 6 36 48; 35 199 4 5 71; 61 81 44 88 9; 85 60 14 25 79] [Si]5x1=[461 277 356 488 393] [Dj]1x5=[278 60 461 116 1060]
25	(Kaur et al., 2018)	N28	3x5	40	[Cij]3x5=[2 7 8 8 3; 5 6 5 5 6; 5 7 8 8 3] [Si]3x1=[2 3 5] [Dj]1x5=[3 1 1 2 3]
26	(Kousalya1 & Malarvizhi, 2016)	N29	3x4	3857	[Cij]3x4 = [15 1 42 33; 80 42 26 81; 90 40 66 60] [Si]3x1 = [23 44 33] [Dj]1x4 = [23 31 16 30]
27	(Ahmed et al., 2015)	N30	3x3	465	[Cij]3x3 = [13 21 14; 8 12 21; 15 17 19] [Si]3x1 = [13 20 5] [Di]1x3 = [12 15 11]
28	(Ji & Chu, 2002)	N31	3x4	1480	[Cij]3x4 = [15 27 13 19; 18 21 24 14; 21 15 16 17] [Si]3x1 = [40 40 20] [Dj]1x4 = [30 20 30 20]
29	(Babu et al., 2020)	N32	4x5	348	[Cij]4x5 = [4 4 9 10 13; 7 9 8 10 4; 9 3 7 10 6; 11 4 10 6 9] [Si]4x1 = [100 90 80 70] [Dj]1x5 = [60 40 90 70 80]
30	(Amreen & Venkateswarlu, 2024)	N33	2x4	151750	[Cij]2x4 = [400 510 800 270; 600 450 750 500] [Si]2x1 = [200 100] [Dj]1x4 = [100 100 75 25]
31	(Sahito, 2021)	N34	4x5	510	[Cij]4x5 = [7 6 4 5 9; 8 5 6 7 8; 6 8 9 6 5; 5 7 7 8 6] [Si]4x1 = [40 30 20 10] [Dj]1x5 = [30 30 15 20 5]
32	(Juman & Hoque, 2015)	N35	3x4	417	[Cij]3x4 = [13 18 30 8; 55 20 25 40; 30 6 50 10] [Si]3x1 = [8 10 11] [Di]1x4 = [4 7 6 12]

2. The synthesis data of 5 sets can be seen in Table 3.2 below.

Table 3.2 Data synthesis

No	TP Name	TP Size	Optimal Value	Matrix
1	S1	5x5	18855	$[C_{ij}]_{5 \times 5} = [27 \ 27 \ 9 \ 33 \ 9;$ $36 \ 15 \ 33 \ 30 \ 15;$ $18 \ 21 \ 27 \ 36 \ 33;$ $33 \ 36 \ 21 \ 24 \ 9;$ $27 \ 9 \ 9 \ 21 \ 15]$ $[S_i]_{5 \times 1} = [150 \ 240 \ 135 \ 300 \ 285]$ $[D_j]_{1 \times 5} = [270 \ 195 \ 210 \ 270 \ 165]$
2	S2	5x5	2790	$[C_{ij}]_{5 \times 5} = [80 \ 80 \ 20 \ 100 \ 20;$ $110 \ 40 \ 100 \ 90 \ 40;$ $50 \ 60 \ 80 \ 110 \ 100;$ $100 \ 110 \ 60 \ 70 \ 20;$ $80 \ 20 \ 20 \ 60 \ 40]$ $[S_i]_{5 \times 1} = [400 \ 700 \ 350 \ 900 \ 850]$ $[D_j]_{1 \times 5} = [800 \ 550 \ 600 \ 800 \ 450]$
3	S3	11x12	1044180	$[C_{ij}]_{11 \times 12} = [1 \ 52 \ 53 \ 54 \ 55 \ 56 \ 57 \ 58 \ 59 \ 60 \ 61 \ 12;$ $63 \ 14 \ 65 \ 66 \ 67 \ 68 \ 69 \ 70 \ 71 \ 72 \ 73 \ 74;$ $75 \ 76 \ 27 \ 78 \ 79 \ 80 \ 81 \ 82 \ 83 \ 84 \ 85 \ 86;$ $87 \ 88 \ 89 \ 40 \ 91 \ 92 \ 93 \ 94 \ 95 \ 96 \ 97 \ 98;$ $99 \ 100 \ 101 \ 102 \ 53 \ 104 \ 105 \ 106 \ 107 \ 108 \ 109 \ 110;$ $111 \ 112 \ 113 \ 114 \ 115 \ 66 \ 117 \ 118 \ 119 \ 120 \ 121 \ 122;$ $123 \ 124 \ 125 \ 126 \ 127 \ 128 \ 79 \ 130 \ 131 \ 132 \ 133 \ 134;$ $135 \ 136 \ 137 \ 138 \ 139 \ 140 \ 141 \ 92 \ 143 \ 144 \ 145 \ 146;$ $147 \ 148 \ 149 \ 150 \ 151 \ 152 \ 153 \ 154 \ 105 \ 156 \ 157 \ 158;$ $159 \ 160 \ 161 \ 162 \ 163 \ 164 \ 165 \ 166 \ 167 \ 118 \ 169 \ 170;$ $171 \ 172 \ 173 \ 174 \ 175 \ 176 \ 177 \ 178 \ 179 \ 180 \ 131 \ 182]$ $[S_i]_{11 \times 1} = [50 \ 60 \ 70 \ 80 \ 90 \ 100 \ 110 \ 120 \ 130 \ 140 \ 310]$ $[D_j]_{1 \times 12} = [50 \ 60 \ 70 \ 80 \ 90 \ 100 \ 110 \ 120 \ 130 \ 140 \ 150 \ 160]$
4	S4	11x12	92212	$[C_{ij}]_{11 \times 12} = [13 \ 35 \ 30 \ 36 \ 35 \ 30 \ 30 \ 33 \ 31 \ 36 \ 15 \ 39;$ $39 \ 30 \ 25 \ 32 \ 28 \ 13 \ 11 \ 32 \ 34 \ 27 \ 14 \ 25;$ $12 \ 39 \ 15 \ 11 \ 13 \ 30 \ 30 \ 31 \ 12 \ 25 \ 32 \ 13;$ $33 \ 10 \ 14 \ 38 \ 38 \ 12 \ 12 \ 15 \ 13 \ 15 \ 37 \ 14;$ $37 \ 11 \ 25 \ 10 \ 14 \ 35 \ 29 \ 14 \ 33 \ 14 \ 37 \ 37;$ $79 \ 88 \ 70 \ 70 \ 83 \ 71 \ 92 \ 93 \ 77 \ 95 \ 82 \ 77;$ $80 \ 98 \ 98 \ 84 \ 73 \ 91 \ 88 \ 86 \ 81 \ 78 \ 73 \ 73;$ $83 \ 94 \ 77 \ 94 \ 76 \ 91 \ 81 \ 70 \ 85 \ 91 \ 99 \ 84;$ $78 \ 75 \ 78 \ 81 \ 79 \ 81 \ 81 \ 98 \ 83 \ 95 \ 88 \ 77;$ $99 \ 81 \ 79 \ 87 \ 89 \ 81 \ 76 \ 74 \ 76 \ 78 \ 89 \ 94;$ $91 \ 83 \ 80 \ 92 \ 97 \ 97 \ 80 \ 89 \ 92 \ 71 \ 83 \ 99]$ $[S_i]_{11 \times 1} = [200 \ 106 \ 139 \ 137 \ 249 \ 171 \ 172 \ 113 \ 230 \ 241 \ 199]$ $[D_j]_{1 \times 12} = [315 \ 309 \ 41 \ 43 \ 110 \ 162 \ 274 \ 184 \ 70 \ 201 \ 125 \ 123]$
5	S5	11x12	116020	$[C_{ij}]_{11 \times 12} = [1 \ 52 \ 53 \ 54 \ 55 \ 56 \ 57 \ 58 \ 59 \ 60 \ 61 \ 12;$ $63 \ 14 \ 65 \ 66 \ 67 \ 68 \ 69 \ 70 \ 71 \ 72 \ 73 \ 74;$ $75 \ 76 \ 27 \ 78 \ 79 \ 80 \ 81 \ 82 \ 83 \ 84 \ 85 \ 86;$ $87 \ 88 \ 89 \ 40 \ 91 \ 92 \ 93 \ 94 \ 95 \ 96 \ 97 \ 98;$ $99 \ 100 \ 101 \ 102 \ 53 \ 104 \ 105 \ 106 \ 107 \ 108 \ 109 \ 110;$ $111 \ 112 \ 113 \ 114 \ 115 \ 66 \ 117 \ 118 \ 119 \ 120 \ 121 \ 122;$ $123 \ 124 \ 125 \ 126 \ 127 \ 128 \ 79 \ 130 \ 131 \ 132 \ 133 \ 134;$ $135 \ 136 \ 137 \ 138 \ 139 \ 140 \ 141 \ 92 \ 143 \ 144 \ 145 \ 146;$ $147 \ 148 \ 149 \ 150 \ 151 \ 152 \ 153 \ 154 \ 105 \ 156 \ 157 \ 158;$ $159 \ 160 \ 161 \ 162 \ 163 \ 164 \ 165 \ 166 \ 167 \ 118 \ 169 \ 170;$ $171 \ 172 \ 173 \ 174 \ 175 \ 176 \ 177 \ 178 \ 179 \ 180 \ 131 \ 182]$ $[S_i]_{11 \times 1} = [50 \ 60 \ 70 \ 80 \ 90 \ 100 \ 110 \ 120 \ 130 \ 140 \ 310]$ $[D_j]_{1 \times 12} = [50 \ 60 \ 70 \ 80 \ 90 \ 100 \ 110 \ 120 \ 130 \ 140 \ 150 \ 160]$

1. Five Real Data of XYZ Companies

NO	Data Real
1	[Cij]2x94 = [22 38 32 17 31 15 23 25 25 20 33 30 18 12 23 18 26 20 15 18 15 25 28 23 16 18 21 28 20 20 25 18 18 18 25 27 21 25 23 26 33 38 29 17 38 17 39 39 22 22 35 25 28 31 33 20 36 31 23 27 25 35 30 31 28 27 28 30 27 54 43 52 56 46 59 53 40 58 44 51 40 43 45 49 48 57 46 46 60 61 66 59 55 56; 26 41 36 21 35 20 27 30 28 25 38 34 23 15 20 15 23 18 14 17 13 23 25 25 17 23 23 33 18 19 28 23 20 18 25 31 20 25 25 23 39 36 28 12 33 14 38 38 20 28 33 28 26 28 31 17 38 30 30 25 22 33 26 29 31 34 28 33 31 49 38 49 52 43 56 49 36 54 40 46 36 39 41 46 44 54 41 43 57 57 62 55 52 51] [Si]2x1 = [308252 138801] [Di]1x94 = [3852 4759 1312 3451 7537 2827 3878 15224 2530 5690 2494 2383 32901 7314 2486 0 249 2735 498 1989 4475 6091 373 5492 17978 4305 23799 1597 2773 2556 48509 3279 4183 1303 795 4024 1381 2597 1666 4848 125 0 0 10192 0 4475 3356 0 2611 2984 1368 0 1119 249 0 9447 2984 2362 125 2984 26599 746 249 0 14055 1437 2283 556 812 5786 2034 5911 5026 524 1133 464 1792 1330 1457 1668 3827 7947 297 759 9074 1436 864 2217 1160 519 4166 614 19405 30392]
2	[Cij]2x94 = [22 38 32 17 31 15 23 25 25 20 33 30 18 12 23 18 26 20 15 18 15 25 28 23 16 18 21 28 20 20 25 18 18 18 25 27 21 25 23 26 33 38 29 17 38 17 39 39 22 22 35 25 28 31 33 20 36 31 23 27 25 35 30 31 28 27 28 30 27 54 43 52 56 46 59 53 40 58 44 51 40 43 45 49 48 57 46 46 60 61 66 59 55 56; 26 41 36 21 35 20 27 30 28 25 38 34 23 15 20 15 23 18 14 17 13 23 25 25 17 23 23 33 18 19 28 23 20 18 25 31 20 25 25 23 39 36 28 12 33 14 38 38 20 28 33 28 26 28 31 17 38 30 30 25 22 33 26 29 31 34 28 33 31 49 38 49 52 43 56 49 36 54 40 46 36 39 41 46 44 54 41 43 57 57 62 55 52 51] [Si]2x1 = [341015 138738] [Di]1x94 = [3332 3976 1252 3494 9871 3027 5214 20029 2437 5584 2615 2184 39154 6641 2735 0 249 2984 746 2238 4848 6837 373 4677 17447 4487 24968 1804 3137 2538 50087 3445 4375 1444 919 4024 1389 2960 1660 5221 125 0 0 11063 0 4972 3729 125 2859 5096 1492 0 871 249 125 10129 3232 2735 249 12554 28961 746 125 0 12664 1382 2405 479 724 6258 1924 5195 5217 536 924 385 2053 1111 1340 1712 3630 8300 321 679 8954 1385 862 2372 1225 480 4369 556 20814 27658]
3	[Cij]2x94 = [22 38 32 17 31 15 23 25 25 20 33 30 18 12 23 18 26 20 15 18 15 25 28 23 16 18 21 28 20 20 25 18 18 18 25 27 21 25 23 26 33 38 29 17 38 17 39 39 22 22 35 25 28 31 33 20 36 31 23 27 25 35 30 31 28 27 28 30 27 54 43 52 56 46 59 53 40 58 44 51 40 43 45 49 48 57 46 46 60 61 66 59 55 56; 26 41 36 21 35 20 27 30 28 25 38 34 23 15 20 15 23 18 14 17 13 23 25 25 17 23 23 33 18 19 28 23 20 18 25 31 20 25 25 23 39 36 28 12 33 14 38 38 20 28 33 28 26 28 31 17 38 30 30 25 22 33 26 29 31 34 28 33 31 49 38 49 52 43 56 49 36 54 40 46 36 39 41 46 44 54 41 43 57 57 62 55 52 51]

	[Si]2x1 = [280110 138801] [Di]1x94 = [3300 4628 1250 2852 8202 2916 3912 16157 2695 5526 2699 2271 41855 7074 1616 0 125 1741 373 1243 2859 3854 249 5151 17117 4368 24291 1560 2679 2952 42223 3585 4182 1423 838 3282 1491 2324 1553 3108 125 0 125 6464 0 2859 2238 0 1616 2921 871 0 498 125 0 6091 1865 1616 125 2921 17029 498 249 0 13885 1516 2480 410 799 6616 1980 5900 4721 584 1032 419 1702 1290 1536 1725 3455 7792 341 632 9100 1269 897 2333 1296 452 3957 487 20317 32228]
4	[Cij]2x94 = [22 38 32 17 31 15 23 25 25 20 33 30 18 12 23 18 26 20 15 18 15 25 28 23 16 18 21 28 20 20 25 18 18 18 25 27 21 25 23 26 33 38 29 17 38 17 39 39 22 22 35 25 28 31 33 20 36 31 23 27 25 35 30 31 28 27 28 30 27 54 43 52 56 46 59 53 40 58 44 51 40 43 45 49 48 57 46 46 60 61 66 59 55 56; 26 41 36 21 35 20 27 30 28 25 38 34 23 15 20 15 23 18 14 17 13 23 25 25 17 23 23 33 18 19 28 23 20 18 25 31 20 25 25 23 39 36 28 12 33 14 38 38 20 28 33 28 26 28 31 17 38 30 30 25 22 33 26 29 31 34 28 33 31 49 38 49 52 43 56 49 36 54 40 46 36 39 41 46 44 54 41 43 57 57 62 55 52 51] [Si]2x1 = [350910 142165] [Di]1x94 = [3512 4300 1291 3477 9394 3031 4597 22500 1776 5384 2824 2530 42638 7341 2735 0 249 2984 746 2238 4848 6837 3737 5858 16806 4210 24717 1715 2549 2520 50440 3185 4359 1437 914 3120 1551 2109 1578 5345 125 0 0 11187 0 4972 3729 125 2859 8452 1492 0 871 249 125 10317 3232 2735 249 8452 29085 746 373 0 13607 1421 2068 554 877 6207 1844 6259 4824 610 1175 403 1840 1110 1389 1739 3546 8113 348 753 8366 1329 970 2121 1181 530 3778 497 19912 34947]
5	[Cij]2x94 = [22 38 32 17 31 15 23 25 25 20 33 30 18 12 23 18 26 20 15 18 15 25 28 23 16 18 21 28 20 20 25 18 18 18 25 27 21 25 23 26 33 38 29 17 38 17 39 39 22 22 35 25 28 31 33 20 36 31 23 27 25 35 30 31 28 27 28 30 27 54 43 52 56 46 59 53 40 58 44 51 40 43 45 49 48 57 46 46 60 61 66 59 55 56; 26 41 36 21 35 20 27 30 28 25 38 34 23 15 20 15 23 18 14 17 13 23 25 25 17 23 23 33 18 19 28 23 20 18 25 31 20 25 25 23 39 36 28 12 33 14 38 38 20 28 33 28 26 28 31 17 38 30 30 25 22 33 26 29 31 34 28 33 31 49 38 49 52 43 56 49 36 54 40 46 36 39 41 46 44 54 41 43 57 57 62 55 52 51] [Si]2x1 = [359746 138801] [Di]1x94 = [3269 4456 1277 2783 8888 3679 4621 18426 2745 5940 2408 2437 48268 5585 3108 0 249 3356 746 2486 5594 7707 373 4944 19086 3758 27005 1884 3139 2275 48013 3411 4479 1386 911 3405 1402 3198 1657 5967 249 0 0 12678 0 5594 4226 125 3232 7955 1616 0 995 373 125 11560 3605 3108 249 7955 32938 871 249 0 13500 1377 2189 589 752 5974 1588 6085 4578 512 1056 452 2240 1138 1268 1818 3634 8274 300 703 8162 1308 901 2438 1088 454 4199 613 19521 29812]

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